

INTERNATIONAL CONFERENCE ON RECENT TRENDS IN ADVANCED COMPUTING
2019, ICRTAC 2019

AI-Based Enhancement of Base Station Handover

V. Kavitha^{*}, G.Manimala, R. Gokul Kannan

Sri Sai Ram Engineering College, Chennai 600044, India

Abstract

With exponential increase in digital data transfer and growing consumer base, there is a need to efficiently handle large set of users. In addition to that, we face a lot of problems due to poor signal connectivity during travel. Though operators look towards plantation of additional base stations or additional hardware, there is also a need to efficiently regulate the existing ones to utilize the resources in an effective manner thereby avoiding band exhaustion and also facilitating the handover of mobile stations in travel. When a mobile device is at travel, the device determines the next base station to which it has to be connected using the Received Signal Strength and requests the same. Therefore, prediction of the next potential base station in advance leads to a hexa-directional ambiguity (the mobile station may move to any of the adjacent 6 hexagonal cells in future which can't be predicted easily in advance). Ping-pong effects during the handovers are also a problem. Few other systems have been introduced which requires tracking of the device over a period of time to determine the trajectory thereby predicting the next potential base station in advance leading to privacy concerns. An addition to the system is proposed that could manipulate these handovers in an efficient manner while conserving privacy. We propose a triangular way of determining the next potential base station in advance. This method eliminates the hexa-directional ambiguity and determines the next one perfectly in advance thereby increasing the efficiency manifold. Reservation of band is done based on the prediction thereby reducing connectivity delays. Improving the existing ones manifold will reduce the requirement of additional hardware thereby facilitating sustainable development by reducing the dangerous impacts on Mother Nature.

© 2019 The Authors. Published by Elsevier B.V.

This is an open access article under the CC BY-NC-ND license (<http://creativecommons.org/licenses/by-nc-nd/4.0/>)

Peer-review under responsibility of the scientific committee of the INTERNATIONAL CONFERENCE ON RECENT TRENDS IN ADVANCED COMPUTING 2019.

Keywords: Hexa-directional ambiguity; ping-pong effect ; received signal strength ; reservation ; tracking ; triangular way;

^{*} V.Kavitha. Tel.: +91 9751237156

E-mail address: kavitha.cse@sairam.edu.in

1. Introduction

Various operators provide mobile connectivity and services to a huge number of users. The increasing requirements for providing connectivity to more number of users with good reliability and higher data rates leads to plantation of more number of base stations or increase in hardware (The stations at which more traffic occurs are monitored now and then to plant supporting base stations or to add hardware to existing ones to increase the capacity of the same). This leads to increased power consumption, environmental hazards and health hazards. A software update is also rolled out now and then to improve the existing ones. In the existing system, there are six base station cells around each base station cell (Hexagonal array of any one operator is considered, where supporting base stations within the same hexagonal cell are considered to be a part of the cell's main base station itself). When a mobile station travels from one region to another it is always surrounded by 6 hexagonal cells. It scans for all the available base stations. The mobile station then decides the next base station to which it can connect through the Received Signal Strength (RSS). Once the mobile station chooses the next base station it requests for connectivity with the new base station along with various other parameters. The new base station checks the authenticity of the mobile station with the network and exchanges the key. The mobile station, when it gets enough signal strength from the new base station terminates the connectivity with the old one (soft hand over - it may also lead to ping-pong effect between available ones).

The goal of this paper is to improve the speed of the handover thereby improving reliability during this handover. Few proposals have already been given by others for enhancement process. Few of them use fuzzy logic with neural networks or AI as in [1], [2], [3], [4]. Few of them use direction and velocity based systems as in [4], [5], [6], [7], [8]. In these cases we need to collect and maintain historical data to track the device over a period of time, which may lead to privacy problems along with more memory and processing power requirements. There is also a need to monitor the device at every instant to keep track of the direction, velocity, angle and other parameters (For finding the direction one must keep track of the travelled path to determine the trajectory of the path). A simple yet efficient triangular method is introduced, where the next potential base station can be accurately determined with less power and memory. Unlike other methods, this method just requires the current location of the mobile station instead of a set of locations thereby conserving privacy (the locations of the device need not be stored continuously for direction of travel). With just the current location of the mobile station we can determine the triangle in which the device is present and hence the next potential base station. Distance considerations are taken during scaling. Reservations are also done in advance which can handle changes dynamically. This triangular method also ensures that existing system can function more efficiently with less addressing requirements similar to that in [9]. Protection of data has been while publishing the data was discussed clearly in [10].

2. Method

The mobile station finds the location of itself through Global Positioning System (GPS) and sends the same to the current base station. Then, the current base station determines the triangle in which the mobile station is moving and sends the same to the base station controller. After this the base station controller determines the next potential base station and informs it to the potential base station and the mobile station. The potential base station reserves the band for the incoming connections in advance. The preference for band reservation is based on "closest mobile station first" policy. On reception of the information by the mobile station, it requests the next potential base station for connectivity.

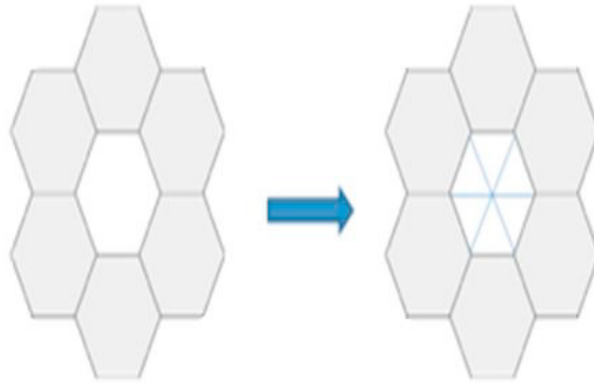


Fig.1. Triangular approach to eliminate hexa-directional ambiguity

If we consider fig.1, we can see a hexagonal array. The central hexagon is the current base station or parent base station region. The hexagonal system is decomposed into triangles. If the system is not decomposed, prediction about next potential base station can't be made in advance without tracking the direction, as the mobile station present in a particular hexagonal region can move to any of the adjacent 6 base station regions in future. This is termed as hexa-directional ambiguity. If direction has to be tracked, it needs storage of locations over a period of time leading to privacy issues. If it is decomposed into triangles, the 2 sides of any triangle fall in the parent base station region (current base station region) leaving the other side of the triangle towards the next potential base station. This helps us to accurately determine the next potential base station in advance. Only the current position is required to determine the triangle and the next potential base station which ensures privacy to greater extent unlike other direction based systems. This also eliminates the requirements for Received Signal Strength (RSS) analysis reducing ping pong effects.

For understanding triangle determination we must consider the 3 lines that split the hexagon into triangles. They are analogous to the line equations:

- (i) $x = y$
- (ii) $x = -y$
- (iii) $y = 0$

The actual line equations are determined from the formula

$$(x - x_1) = \left[\frac{y - y_1}{y_2 - y_1} \right] * (x_2 - x_1) \quad (1)$$

Where (x_1, y_1) and (x_2, y_2) are the end points of the line considered. The end points of the line are GPS coordinates (latitude(W,E) converted to $(-x, x)$ and longitude(N,S) converted to $(-y, y)$) of the vertices of the hexagon in that line, with an adjustment factor for sectional overlaps (The base station with lesser historical traffic and more historical coverage-service efficiency has to be given the preference during adjustment).

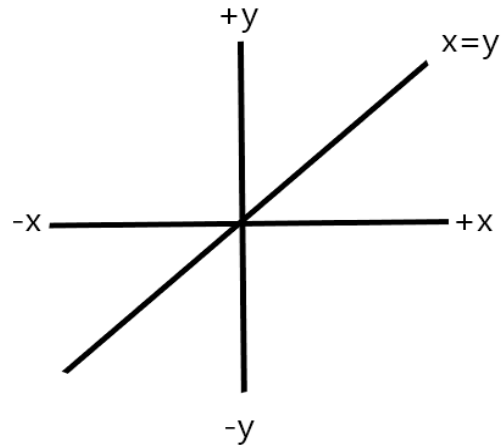


Fig.2 Graph of line $x = y$ (similar to one of the lines that splits the hexagon into triangles)

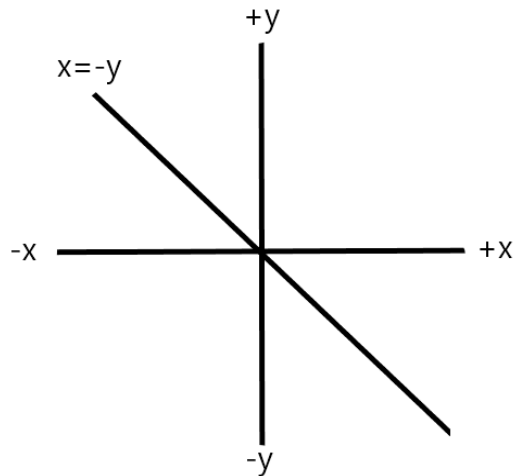


Fig.3 Graph of line $x = -y$ (similar to one of the lines that splits the hexagon into triangles)

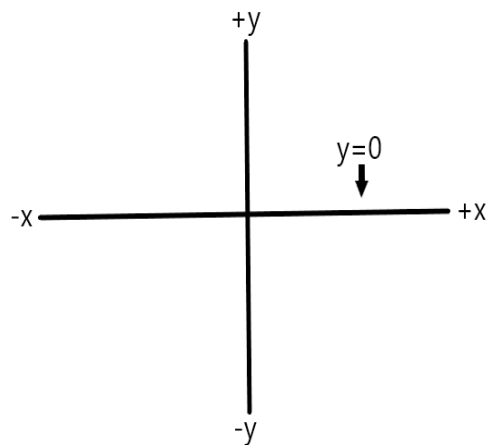


Fig.4 Graph with representation for $y = 0$ (similar to one of the lines that splits the hexagon into triangles) For each type “(i)”, “(ii)” and “(iii)”, we must apply the end points in “(1)”. After substituting the end points in “(1)”, we will have another set of 3 equations one for each type. Let us term them as “(1A)”, “(1B)” and “(1C)” respectively.

If we consider the equation “(1A)” and the line in the hexagon similar to that in fig.2, we can see that any point (x, y)

- (i) to the left of the line will yield negative value when substituted in “(1A)”
- (ii) to the right of the line will yield positive value when substituted in “(1A)”
- (iii) on the line will yield zero value when substituted in “(1A)”

If we consider the equation “(1B)” and the line in the hexagon similar to that in fig.3, we can see that any point (x, y)

- (i) to the left of the line will yield negative value when substituted in “(1B)”
- (ii) to the right of the line will yield positive value when substituted in “(1B)”
- (iii) on the line will yield zero value when substituted in “(1B)”

If we consider the equation “(1C)” and the line in the hexagon similar to that in fig.4, we can see that any point (x, y)

- (i) above the line will have greater y value than the one obtained in “(1C)”
- (ii) below the line will have lesser y value than the one obtained in “(1C)”
- (iii) on the line will have the same y value to the one obtained in “(1C)”

With this setup one can find the triangle in which the mobile station is present. For finding it, the location is to be applied in “(1A)” and “(1B)”. It is also compared with “(1C)” and the results are noted.

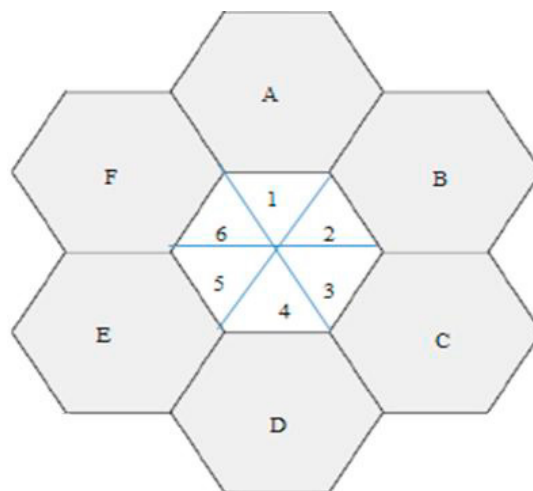


Fig.5. Triangle determination

For example, in fig.5, let's consider a mobile station in triangle 1. When we feed the position of the mobile station in the equations “(1A)” and “(1B)”, we get 1st result as negative (left) and 2nd result as positive (right). On comparing the y value of the mobile station with y value obtained in “(1C)”, it is greater (above). With this one can say the mobile station is at triangle 1 and next potential base station is A.

Considering another case where the mobile station moves from 1 to 6 to 5 to E's hexagon, decoding it stage by stage we get potential base stations as A, F, E. In these cases there are reservations done at A, F and E at each stage respectively. The actual reservation that is needed is at E. To avoid unwanted bandwidth wastage by moving the data of mobile stations here and there, the reservations are alone done with basic information whereas, the data of the mobile station is not moved until registration is done at the new base station. Reservation at base stations A and F may lead to band unavailability for the most needy mobile stations closer to A and F than the considered

mobile station. This is overcome by giving preferences in reservation, which is later mentioned in fig.7. Considering overall performance, the new form of triangle based estimation will give an upper hand.



Fig.6. Ambiguities that may occur

With reference to fig.6, the triangular regions are just representations and the lines are imaginary. There is no such case that the mobile station's location falls on any of the lines leading to ambiguity. It is because, here ranges are considered and therefore mobile stations tend to fall significantly in any one of the ranges. Also the case of superimposition of the mobile station with the base station can't occur because it is the location of the parent base station itself. There is no such chance that the base station device's centre point will superimpose exactly with the mobile station's location when looked in detail. Even if it happens, it means it's at the exact location of the parent base station and is safely connected with the same. There is no need for reservation at this stage. It can be done once it starts moving away from it.



Fig.7. Multiple incoming mobile stations

When there are multiple incoming mobile stations like in fig.7, distance is calculated for each mobile station from the potential base station using the distance formula

$$D = \text{sqrt} ((x_2 - x_1)^2 + (y_2 - y_1)^2) \quad (2)$$

Where (x_1, y_1) and (x_2, y_2) are the locations of the mobile station and the potential base station respectively. The priority is based on the "closest mobile station first policy". From fig.7, we can see that, the band for the mobile station that is closer (red) is reserved first followed by the other (green).

3. Conclusion and Future Scope

AI is of different types. If we look at the lowest levels, any AI system is a set of if-else conditions. This is a rule based AI system with the defined logic that has ability to adapt to different cases. The entire system can also be considered as Internet of Things as the entities communicate over the network (Internet when scaled) to perform the function required. The communications between these entities are modular with each one considered as an object, thereby improving security, scalability and power efficiency. This also helps in reducing the addressing space requirements by localizing the addresses with subnets. The delay during the request of connection by mobile station to next potential base station is reduced significantly. When there are multiple operators or multiple additional base stations in a region to support traffic, the mobile station knows exactly which one to request for connection instead of Received Signal Strength analysis, thereby reducing delay considerably. Ping-pong effects are mitigated. The accepting base station can also accept the incoming request quickly as it already has the information of the incoming one and has already made a reservation. The data about the mobile station is not transferred completely to the potential base station until registration is done for bandwidth conservation as mentioned earlier. On comparison with the RSS system, location-direction based systems and other AI based systems, this system tend to have less memory, power and processing requirements (Need not store and process all the historical locations at each stage). It also promises good efficiency along with good amount of privacy and scalability.

Multiple other parameters such as city traffic flow, urbanization index, data flow index etc. can be used to extend the model to have further enhancements. Complex multi-functional AI can serve complete automation and reinforcement based on dynamic needs. For such a task, historical data from the operator's side is required.

Acknowledgement

We thank all of them mentioned in the references, for putting forward significant efforts which has helped us to a greater extent. We also thank each and everyone, who have put forward their efforts in this domain.

References

1. Thumthawatworn, Thanachai & Pervez, Anjum & Santiprabhob, Pratit. (2013). "Adaptive Traffic Dependent Fuzzy-based Vertical Handover for Wireless Mobile Networks". Conference: ICN 2013, The Twelfth International Conference on Networks.
2. Aibinu, A & Onumanyi, Adeiza & Adedigba, Adeyinka & Ipinoyomi, Michael & Folorunso, Taliha & Salami, Momoh. (2017). "Development of hybrid artificial intelligent based handover decision algorithm." *Engineering Science and Technology, an International Journal*. 20. 10.1016/j.jestech.2017.01.005.
3. Muñoz, P & Barco, Raquel & de la Bandera Cascales, Isabel. (2014). "Load balancing and handover joint optimization in LTE networks using Fuzzy Logic and Reinforcement Learning. *Computer Networks*." 76. 10.1016/j.comnet.2014.10.027.
4. Monil, Mohammad Alaul Haque & Qasim, Romasa & Rahman, Mohammad. (2013). "Speed and direction based fuzzy handover system." *IEEE International Conference on Fuzzy Systems*. 1-8. 10.1109/FUZZ-IEEE.2013.6622560.
5. Akyildiz, Ian & Ho, J.S.M. & Lin, Yi-Bing. (1996). "Movement-Based Location Update and Selective Paging for PCS Networks." *Networking, IEEE/ACM Transactions on*. 4. 629 - 638. 10.1109/90.532871.
6. Li, Jie & Kameda, Hisao & Li, KQ. (2000). "Optimal dynamic mobility management for PCS networks." *Networking, IEEE/ACM Transactions on*. 8. 319-327. 10.1109/90.851978.
7. Chen, M.-M & Yang, Yan & Zhong, Z.-D. (2015). "Location-Based Handover Decision Algorithm in LTE Networks under High-Speed Mobility Scenario." 2015. 10.1109/VTCSpring.2014.7022977.
8. Ben-Mubarak, Mohammed & Ali, Borhanuddin & Noordin, Nor & Ismail, Alyani & Ng, Chee Kyun. (2011). "Mobile Station Movement Direction Prediction (MMDP) Based Handover Scanning for Mobile WiMAX System. *Wireless Personal Communications*." 73. 10.1007/s11277-013-1220-y.
9. Chepoi, V., Dragan, F.F. & Vaxès, Y. "Addressing, distances and routing in triangular systems with applications in cellular networks." *Wireless Netw* (2006) 12: 671. <https://doi.org/10.1007/s11276-006-6527-0>.
10. Nithya M & co., "Predictive delimiter for multiple sensitive attribute publishing." *Cluster Computing* (2019) 22(Suppl 5): 12297. <https://doi.org/10.1007/s10586-017-1612-y>